GROVE - Green Robotics Orchestration and Value Ecosystems

Project proposal for upcoming Vinnova calls

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GROVE - Green Robotics Orchestration and Value Ecosystems

Autonomous robot fleets enable
future smart outdoor infrastructure.

Robot-generated data enables:

predictive maintenance,
operational optimization,
surveillance & safety services.

Distributed IoT-edge intelligence
creates new business opportunities.

Key message: From autonomous mowers → connected service ecosystems.

Joint Initiative: Husqvarna + LTU

Husqvarna

- Business development
- New digital services
- Ownership, leasing & servitization models

LTU

- ***GROVE Arena*** cyber-physical experimentation environment
- Distributed analytics & inference
- Decentralized orchestration
- Simulated infrastructure, combined with one or a few real devices

Goal: Experimentally demonstrate future distributed robot-service ecosystems — not replace existing production systems.

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Demonstrations & infrastructure

Husqvarna Group

Business values

GROVE Arena: Experimental Cyber-Physical Environment

Onboard Mowers

- Local sensing & AI inference
- Mower-to-mower communication
- Real robotic experimentation platforms

Edge Nodes

- EPOS/UWB/WiFi infrastructure
- Edge servers & compute resources
- Aggregation & orchestration
- Cooperative analytics

Infrastructure

- Real robotic devices
- Edge-computing hardware
- Communication & positioning systems
- Large-scale simulation & emulation

Key challenge: Integrated decentralized operational intelligence across physical and simulated infrastructures.

Deliverables & Impact: Demonstrators & Business Value

Deliverables

- GROVE Arena experimentation environment
- Distributed analytics & inference pipelines
- Operational intelligence dashboard
- Ad hoc cooperative analytics

Business Demonstrators

- Reduced administrative overhead
- Improved service quality
- Automated maintenance coordination
- New digital services

Evaluation

- Real robotic platforms & edge hardware
- Large-scale simulation & emulation
- Integrated cyber-physical experimentation

Impact: Future digital ecosystems for autonomous outdoor robots.

Contact

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